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Networks in Meetings: How do People connect?

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Abstract

Meetings are not static events but rather dynamic in nature. The course of the meeting is affected by participants' responses to one another. In this chapter, we argue that these responses among meeting participants can be understood as ties in an interaction network. This network perspective to meeting interaction allows us to focus on the relationships and connections between meeting participants who are embedded in a web of interrelations. Social network analysis helps us to systematically uncover these connections. By applying social network analysis to meeting interaction, we take the relational nature of social interaction into account and use a structural perspective to meeting interaction. We introduce the central principles of the network perspective and explain how meetings can be conceptualized as social networks. Next, we focus on typical social network measures and explore how these network measures can be applied to meeting interaction. We further provide a detailed example to showcase how social network analysis can be applied to real team meeting data. We close our chapter by discussing benefits and challenges of using social network analysis in team meetings.

Keywords: Social network analysis; interaction process analysis; meeting behaviors

Introduction

Meetings are an increasingly important part of organizational life (e.g., Cohen, Rogelberg, Allen, & Luong, 2011; Rogelberg, Leach, Warr, &Burnfield, 2006) and spark both researchers' and practitioners' interest. Frequent activities in team meeting encompass sharing knowledge, solving problems, and making decisions (e.g., Kauffeld & Lehmann-Willenbrock, 2012). All of these activities require interaction among meeting participants. For example, the meeting leader opens the meeting and introduces the topic of today's meeting. Next, meeting attendees are asked to provide their knowledge on the subject. They might exchange opinions, encourage each other's participation, make plans, way pros and cons of certain solutions, and decide on actions to be carried out after the meeting. They might also get lost in details, gossip or complain about the situation (e.g., Kauffeld & Meyers, 2009). In doing so, some meeting participants might engage more strongly in the discussion, provide more ideas, and share a bigger part of the overall conversation while other meeting participants might be relatively silent and share less speaking turns. All in all, a constant stream of interaction among meeting participants emerges as the meeting progresses. Because meetings are not static events, this interaction is dynamic in nature. A participant's action influences the reactions of the other meeting participants and affects the course of the meeting (Bonito, 2002). So, how can we capture such dynamics among meeting participants?

One method that is suitable for such a task is social network analysis. In the most abstract definition, a social network is defined as a set of *nodes* (i.e., actors) connected by a set of *ties* representing a specific relationship (Wasserman & Faust, 1994). Instead of individuals' attributes, the network perspective focuses on these relationships between actors. It emanates from the idea that in all social interactions actors are embedded in a web of interrelations, which connects them with other actors and defines their individual roles in the social structure. These network structures provide opportunities and constraints for each actor

and have a strong influence on interaction processes in social situations (Brass, 2012). Focusing on team meeting processes and structural aspects of participation, social network analysis thus allows us to capture the distribution of group participation over the course of a meeting, to identify key positions of team members, and to gain insights into the interaction structure between meeting participants. For example, we can picture who is connected to whom and which meeting participant plays the most central role during the meeting.

In this chapter, we will first introduce the tenets of social network analysis and explain how team meetings can be conceptualized as social networks. Next, we will describe different social network measures and report central research findings. Moreover, we will present an application sample of interaction networks in team meetings to explore factors that are related to the overall interaction structure in meetings. Finally, we discuss benefits and challenges of using social network analysis in team meetings and outline future research questions that still need to be answered.

Meetings as Networks

Imagine two team meetings that might happen right at this moment in an organization. Both discuss current work problems but take completely different courses of action. In Team 1, Steve (A) starts the discussion with a suggestion of how to proceed. John (B) addresses the current problem of operating a new machine properly. Steve (A) carries on describing the problems in more detail. Brian (C) voices his negative opinion about the new machine but is cut off by John (B). Steve (A) and John (B) start debating the pros and cons of the machine. The discussion goes back and forth between them while the other five team members keep in the background. Mary (D) and Dave (E) side with Steve (A) and occasionally support his suggestions. Brian (C) and Linda (F) are on John’s side and request a couple of explanations from Steve (A). Jenny (G) sides with neither of them. Overall, Brian (C), Mary (D), Dave (E),

Linda (F), and Jenny (G) barely participate in the discussion at all. Over the course of the meeting, the interaction focuses largely on Steve (A) and John (B).

In Team 2, Tom (A) starts the meeting by asking the other team members about current problems. Lisa (B) addresses a problem with their work process. As Rick (C) has been working in the organization the longest; he has the most experience with work practices and thus describes the problem in more detail. Lisa (D) clarifies the contributions and keeps the team focused on the topic. Susan (E) makes a suggestion about the further procedure, acting as the facilitator of the discussion. Gary (F) is the newest team member who has worked at a different assembly line before and thus has knowledge about the work processes at the other line. As Nancy (G) has worked for a different electrical industry company before joining the team, she knows how similar problems have been solved in the other organization. During the entire meeting, all team members are actively involved in the meeting process of discussing the problem and finding solutions.

We argue that the conversational responses between meeting participant represent the ties that connect participants as the responder reacts to the previous statement. This verbal interchange between participants thus shapes the structure of the meeting interaction network. The examples from Team 1 and Team 2 illustrate how the meeting process can result in completely different patterns of interaction. In Team 1, Steve (A) and John (B) occupy central positions in the meeting, which results in a highly centralized interaction structure. By contrast, the social network in Team 2 is highly decentralized. The interaction networks of the two seven-member teams are contrasted in Figure 1 and Figure 2.

Insert Figure 1 and Figure 2 about here

The idea of conceptualizing meeting interaction as a network is a new concept that offers a wide range of opportunities. Meeting researcher interested in capturing such network structures need to record the sequence of speaking-turns throughout the meeting. We advise to either use audio or video recordings of team meetings as data source. Benefits and downsides of using audio or video recordings as well as an extensive guideline for how to gather such behavioral process data in organizations is discussed in Meinecke and Lehmann-Willenbrock in this volume.

Social Network Perspective

In order to understand how to analyze interaction networks in team meetings, we will give a brief primer on the central principles of the network perspective. In general and as mentioned above, a network is defined as a set of actors (“nodes”) and the relations (“ties”) between them (Wasserman & Faust, 1994). Nodes can represent individuals, groups, organizations, or even societies (Borgatti, Mehra, Brass, & Labianca, 2009). Ties can represent an even broader array of constructs, as depicted in Table 1, including similarities (e.g., same location, same attributes, or group membership), formal relations (e.g., kinship or hierarchy), affective relations (e.g., friendship or trust), cognitive relations (e.g., in the know), interaction (e.g., communication or advice), or flows (e.g., information or resources). Beyond the content, ties can vary in direction (non directional such as being a member of a group or directional such as giving advice), value (binary such as being teammates or valued such as being friends on a scale of one to ten), frequency (e.g., weekly vs. monthly meetings), and medium (e.g., face-to-face contact vs. mail contact). Studying team meetings, participants represent the nodes and their interaction represents the links among them. An overview of relational social network measures of ties is presented in Table 2.

Insert Table 1 and Table 2 about here

Another crucial distinction is made by network researchers regarding the strength of ties (Granovetter, 1973, 1983). Researchers distinguish between strong and weak ties, which have different benefits in different social situations. Strong ties are often seen in relationships to family and friends, whereas weak ties are seen in connections with acquaintances. The benefit of strong ties mostly lies in socio-emotional support, which entails trust and friendship. By contrast, the benefit of weak ties is seen in the diverse, non-redundant information that actors gain from contacts outside their regular circle of friends. Additionally, a vital distinction is made between positive and negative ties (Labianca & Brass, 2006). The positive aspects of social relationships focus on benefits actors gain by their contacts, which create social capital. Social capital is defined as an accumulation of opportunities due to different social contacts (Brass, 2012). Positive relations such as friendship can facilitate knowledge transfer, and in the process, foster competitive success. By contrast, the negative aspects relate to the liabilities of social relationships. Negative relations such as dislike can impede critical information exchange and thereby hamper performance.

Although the network approach spans disciplines including sociology, mathematics, psychology, economics, communication, or political sciences, there are certain fundamental principles that provide unity (Wellman, 1988). First, human behavior is best predicted by examining the web of interrelations in which social actors are embedded. Second, analyses focus on the relationships between actors instead of their specific characteristics. Third, populations are defined relationally not categorically. Thus, interdependence between actors has to be assumed. Fourth, a social structure is not seen as the aggregation of dyadic ties. The embeddedness of the actors in the network influences all interactions, not simply the relation between the two actors engaged in direct interaction. Fifth, boundaries between groups can be

fuzzy rather than firm. Networks with different content can overlap, and actors can have cross-cutting connections to multiple groups.

Social Network Measures

"The structural focus is at the heart of social network analysis, and the abstract nature of patterns of nodes and ties have led to the wide application of networks to a variety of different disciplines" (Brass, 2012, p. 671). The focus of social network analysis is based on three different research approaches: (1) identifying structural positions in the network, (2) detecting subgroups, and (3) describing the overall network. Associated with each focus are different network measures (Kilduff & Brass, 2010).In the following, we will give a short overview of each research approach, provide information on important network measures, and explain how these different social network measures relate to the study of meetings.

Structural positions in networks

No matter how much experience anybody has with social network analysis, simply looking at a sociogram gives us a sense of the influence and power of individual actors based on the network patterns and structures without even asking what the nodes and ties represent (Brass, 2012). Table 3 gives a summary of different social network measures for individual actors.

Insert Table 3 about here

The most frequently used metric to identify structural positions is called *centrality* and describes how much the position contributes to the actor's importance. For example in Figure 1, we identified Steve (A) and John (B) as the central actors. Different aspects of influence are pooled under the term *centrality* such as *degree* (actor's number of direct links to others) which describes an actor's direct influence, *betweeness* (extent to which an actor mediates

between others) which captures an actor's control over the information flow, and *closeness* (extent to which an actor reaches all others) which measures how far information takes to reach the actor. From a meeting perspective, researchers have to distinguish between *in-* and *outdegree* ties. O*utdegrees* are ties that are directed away from an actor whereas *indegrees* are ties that are directed towards an actor. In team meetings, *outdegree* describes how many turns a speaker directs to others. Thus, it captures how many times an actor addresses other participants. *Indegree* captures how many times an actor responds to a previous utterance.

In team meetings, *betweeness* measures how much an actor mediates interchange between other participants. For example, a meeting participant with a high *betweeness* would frequently mediate between other meeting participants, facilitating flow of information. *Closeness* captures how much an actor interacts with all the other meeting participants.

A network measure closely related to *centrality* is *assortativity*. It captures an actor's preference to connect with similar actors. In meetings, this preference can be measured as the tendency of a central actor to interact with other central participants. Beyond *centrality*, network researchers are frequently interested in the social capital an actor draws from his structural position. This is captured by the concept of *structural holes* (e.g., Burt, 1992). *Structural holes* emerge when ties are absent in the network. Actors gain influence by filling these *structural holes* and thereby linking unconnected groups. *Structural holes* can be measured by the *effective size* of the network, which captures the redundancy in the actor's network. A low network constraint is achieved when an actor has ties to others who, in turn, are not interconnected with each other. A low network constraint thus indicates how much the actor bridges *structural holes*. In regard to meeting interaction, *effective size* captures how much the meeting is focused on the dyadic interaction of specific actors.

Subgroups in networks

As all actors are interdependent, the analysis of subsets within the network is a key to understanding network structures (Scott, 2000). Social network measures for subgroups are shown in Table 4.

Insert Table 4 about here

A simple classification of subgroups is achieved by identifying the *components* of a network. *Components* describe groups in which all nodes are either directly or indirectly connected with each other. A relational method of grouping actors is called cohesion. Actors can also be clustered in *cliques* based on their direct ties. A *clique* is a complete subgroup in which everybody is connected to each other. These criteria can be relaxed, however, by a required number of ties, which offers an opportunity for meeting researchers to analyze interaction patterns in meetings such as subgroups of heavily interacting participants. A structural method of grouping actors is called *structural equivalence*. Actors are clustered based on similar patterns of ties. Actors thus occupy similar positions in the structure without necessarily being directly linked. In meetings, subgroups with equivalent interaction patterns might be found such as triads of participants discussing a topic among themselves but not with others.

Overall structure of networks

Finally, several network metrics have been developed to measure the structure of the entire network, which is of special interest for small group research (Katz, Lazer, Arrow, & Contractor, 2004). Table 5 provides an overview of the most commonly used social network measures for entire networks.

Insert Table 5 about here

A basic social network measure for entire networks is *centralization*. It describes the degree to which a network revolves around a single actor. From a meeting perspective, it captures the degree to which all participants are involved in the meeting interaction. *Distance* quantifies the cohesion within the network by capturing the length of links between actors. In meetings, it measures the probability of two participants interacting with each other. *Core-periphery models* describe how much a network is structured around a core with closely linked members and a periphery with members who are only loosely connected to the core. Transferred to team meetings, it measures how much the interaction is focused on a core of heavily interacting participants and a periphery of detached participants. Last but not least, t*ransitivity* quantifies the extent to which connected actors are linked to the same others. From a meeting standpoint, it captures how much dyadic and triadic interaction occurs respectively.

Research on Social Networks

Social network analysis has its roots in research by Moreno (1934, 1951) who is often described as one of the founders of social network analysis (e.g., Wasserman& Faust, 1994). In his early work, Moreno frequently studied friendship networks (e.g., who likes whom and who dislikes whom).As early as the 1950s, researchers started examining interaction structures in small groups (Bavelas, 1950; Leavitt, 1951). In laboratory studies, the effect of specific interaction patterns on group performance was measured. These early studies showed that the complexity of the task plays a key role. With easy tasks centralized structures foster group functioning whereas decentralized structures lead to better group performance when problems are complex. A decentralized structure also produced better results when information was ambiguous or distributed unevenly among the group members (Shaw, 1964, 1971).

Experiencing a resurrection in the 1990s, researchers started taking the investigations on interaction patterns to the field to analyze which structures affect group outcomes in organizational teams. Findings showed that real work groups completing complex tasks such as product development or service improvement perform better with decentralized structures and outperform groups with centralized patterns (Cummings & Cross, 2003; Sparrowe, Liden, Wayne, & Kraimer, 2001; see also Huang & Cummings, 2011). Furthermore, a higher density of interaction within work groups relates to higher productivity (Reagans & Zuckerman, 2001; Reagans, Zuckerman, & McEvily, 2004). Balkundi and Harrison (2006) conducted a meta-analysis showing that density within teams is linked to various measures of team performance. All in all, it can be summarized that work groups benefit from internal cohesion. Lipman-Blumen and Leavitt (2001) applied these findings to their theory of high-performing "hot groups". They defined a "hot group" as a lively, dedicated, and usually small group, whose members fulfill challenging tasks in a quick manner, and argued that fluid, decentralized communication structures are necessary for "hot groups" to be successful (Lipman-Blumen& Leavitt, 2001).

Another stream of research addresses the question, which conditions lead to the emergence of social relations in networks. A well-known effect is called *homophily* which describes people's preference to interact with similar others. It facilitates communication, fosters reciprocity, increases frequency of interaction, and improves trust (Brass, 2012). Several researchers found that employees in organizations were often segregated into two groups by gender. Men tend to have stronger homophilous ties across multiple social relations whereas women tend to have social support ties (e.g., friendship network) to other women but instrumental ties (e.g., advice network) to men (Brass, 1985; Ibarra, 1992).Beyond gender, Brass and Burkhard (1992) demonstrated that status affects homophily as interaction patterns in organizations are often related to similarity in power. Moreover, employees with high

education and low neuroticism tend to occupy central positions in friendship and advice networks (Klein, Lim, Saltz, & Mayer, 2004). People who are individualistic and have a high locus of control tend to bridge structural holes (Kalish & Robins, 2006). Studies on stress in teams as an antecedent of social network structures also yielded interesting insights. Argote, Turner, and Fichman (1989) found that stress in teams fosters the development of centralized structures and team members high in interpersonal dominance emerge to be more central in the communication network (Brown & Miller, 2000). These findings show that a variety of antecedents affect social networks. However, research on the emergence of interaction patterns in teams is still limited and additional antecedents remain to be explored.

As the focus of social network analysis is on structural aspects, subgroup formation has received a great deal of attention. Research on subgroups is often entangled with the concept of faultlines. A faultline divides the members of a group on the basis of one or more attributes. Several studies have shown that subgroups, created by faultlines, strongly influence team processes and outcomes (e.g., Lau &Murningham, 1998; Lau &Murnighan, 2005; Thatcher & Patel, 2011). Group members in strong-faultline subgroups tend to increase the social interaction within their own subgroups and decrease their communication with other subgroups, which reduces cohesion (Mollemann, 2005) and team satisfaction (Thatcher & Patel, 2011).In addition, Markovsky and colleagues found that exclusive cliques disrupt the unity and degrade overall solidarity in teams (Markovsky & Chaffee, 1995; Markovsky & Lawler, 1994). Recent research suggests that the type of subgroup might be a determining factor for the influence of subgroups on team outcomes (e.g., Bezrukova, Jehn, Zanutto, & Thatcher, 2009; Carton & Cummings, 2013). The specific type of a subgroup is determined by the nature of the faultline that formed the subgroup (Bezrukova et al., 2009). Carton and Cummings (2013) considered identity-based and knowledge-based subgroups. In identity-based subgroups, members share the same social category or values. Knowledge-based

subgroups, on the other hand, are formed because of differences in technical languages or approaches to problem solving. While identification-based subgroups mostly hurt team performance, knowledge-based subgroups can enhance productivity (Carton & Cummings, 2013). Labianca and Brass (2006) argue that the social relations within subgroups differ largely from the relationships across subgroups. The social network within subgroups is characterized bya web of positive relations whereas the network across subgroups is characterized by negative relations. These findings suggest that the impact of subgroup formation in teams may influence a whole set of social relations, including the interaction in team meetings.

Beyond the internal interaction in small groups, researchers have explored the context in which work groups are embedded (Ancona & Caldwell, 1992; Stohl & Putnam, 2003). Research on external ties has focused on the relationships linking teams with strategic partners in their environment (Ancona & Caldwell, 1992). Studies on interlocking directorates have shown that these external partners can be crucial for knowledge gain and team success (Haunschild, 1993; Mizruchi, 1996; Oh, Chung, &Labianca, 2004). However, Sparrowe and colleagues (2001) found that a team's external ties were harmful for their performance. Scholars thus argue that several factors have an influence on external ties being beneficial or harmful, including the team's autonomy, their need for external information, and task overload (Baldwin, Bedell, & Johnson, 1997; Haas, 2002). The strength of ties is another influencing factor as weak external ties have been found to be useful to bring in simple information whereas strong external ties are helpful in acquiring complex information (Hansen, 1999).

More recently, researchers have begun to use social network analysis for understanding team leadership (e.g., Mehra, Smith, Dixon, & Robertson, 2006). First results show that teams with a high leader centrality achieve higher team performance (Balkundi &

Harrison, 2006). Looking at leaders' individual outcomes, results show that the position in informal networks plays a crucial role for leaders as it correlates with career success. Leaders who occupy a broker position achieve more promotions and pay raises (Burt, 2004; 2005). A central position in friendship networks relates to personal reputation and group performance (Mehra et al., 2006). Moreover, Burt and Ronchi (2007) demonstrated that making leaders aware of network concepts improves their career success. Despite these first promising findings, however, research on leadership networks is still limited and just beginning to be explored (Hoppe &Reinelt, 2010).

In a new development, a number of researchers have begun to conceptualize groups as complex systems integrating sets of multiple relations between persons, tasks, and tools (Katz et al., 2004). Crawford and LePine (2013) developed a model in which taskwork and teamwork are positioned as alternative network relations that connect team members. Working on the same task or sharing a tool creates a taskwork tie, whereas interacting to set goals, plan further actions, coordinate efforts, or provide support creates a teamwork tie. Reviewing previous work on task- and teamwork, Crawford and LePine (2013) found that different interaction structures such as centralization and subgroup formation can have both positive and negative effects on team effectiveness. As a result, they argue that analyzing the network structure of task- and teamwork interaction provides new insights into team dynamics that are difficult to explore otherwise. Therefore, their integrative approach offers opportunities to gain more complete and extensive explanations of team effectiveness and functioning.

Each of these independent lines of research has helped us understanding features of teams that are inherently structural. The findings yield important insights into communication and interaction structures in small groups and offer interesting implications for meeting research as well. Research on network antecedents and consequences has provided large

support for the influence of homophily and cohesion. As similarity eases communication, individual characteristics of meeting participants might also influence the frequency of dyadic interaction, emergence of interaction patterns, and formation of subgroups. Density of communication and decentralized interaction structures might be important for meeting functioning and meeting satisfaction. Focusing on individual positions in networks, scholars have the opportunity to explore meeting leadership and participants' roles in meetings, including facilitators and specialized experts. Integrating research on faultlines and social networks, scholars might gain fruitful information on the influence of subgroups in meetings. Examining participants' external ties offers the advantage of accounting for the environment and meeting context. Ties to key external partners might provide a meeting participant with useful information and influence his or her position in the interaction network. A dense external network might increase the team's organizational knowledge and result in decentralized interaction networks in meetings. Finally, the approach of multiplex ties (see Table 2) opens interesting research vistas. Multiple social relations might be associated with interaction ties. Participants connected in the team's social support network might also interact more frequently in a meeting.

To give a deeper insight into the application of social network analysis on meeting data, we will present a study by Sauer and Kauffeld (2013) that specifically focuses on the overall interaction structure of team meetings to showcase how centralization can be measured based on meeting participant's verbal interactions.

Social Network Analysis in Team Meetings: An Application

The aim of the study by Sauer and Kauffeld (2013) was to analyze naturally occurring social networks in team meetings. Specifically, the study focused on testing a centralization measure of meeting interaction and examining its effect on team outcomes. Therefore, data was gathered in a field study in two medium-sized German companies. The sample comprised

a total of 332 employees nested in 54 teams. Participants were videotaped during regular team meetings. The attendees were advised to ignore the camera and talk like they would normally do. During the meeting, participants discussed current work-related problems and their solutions. At the end of each meeting, the attendees were asked to rate their perceived team performance using a scale by Kirkman and Rosen (1999).

For analysis of the videotaped meeting interaction data, speaking-turns were coded by a pool of five trained coders using Interact software (Mangold, 2010). Every single turn of talk was directly cut from the video stream. On average, 3.6% of speaking-turns were turns with meeting participants speaking at the same time. These speaking-turns were not included in the subsequent analysis of meeting networks. Thus, turn-taking was very orderly overall (see also Dabbs & Ruback, 1987).

To structure all data, speaking-turns for each meeting participant were counted and fed into a “who-to-whom” matrix that displays the frequency of responses of each speaker to others (i.e., cell scores), the number of responses a speaker receives (i.e., indegree; column total), and the number of turns a speaker directed at others (i.e., outdegree; row total). The sum of indegrees and outdegrees has to be the same and represents the overall number of speaking-turns in the team meeting. Next, group participation (i.e., the number of speaking-turns for each meeting participants as a proportion of the group total of speaking-turns) and participation equality were calculated. Participation equality is expressed as the standard deviation of group participation across all team members and indicates the extent to which participation of all meeting participant is equally distributed across the team (Paletz & Schunn, 2011). A high participation equality score indicates a high heterogeneity in group participation. Finally, interaction centralization was calculated according to Freeman (1979). The centralization index provides information on the structural distribution of group participation and shows how strong interaction is concentrated on one team member (Brass,

2012). In doing so, the network centralization of a given network was compared to the maximal possible central interaction structure (for a detailed description see Sauer & Kauffeld, 2013). The network with maximal possible centralization is a star network. In this case the central actor is the only actor who has ties to all other team members while they are not directly connected with one another.

Meeting centralization ranged from 0.10 to 0.69 with a mean of 0.43 and a standard deviation of 0.16. Meeting centralization was not significantly correlated with team meeting size. Multilevel regression analysis was applied to link meeting centralization to perceived team performance. The advantage of this approach is that it considers the hierarchical structure of the data (individuals nested in teams). As control variables, gender and age were inserted in the analysis on the individual level and group size on the group level. Three models were compared with model 1 comprising gender, age, and individual participation. Participation equality was added to model 2 and meeting centralization to model 3 as a group level predictor. The chi-square difference test with scaling correction factors was used to compare the fit of the models. Results showed a significant difference between model 2 and 3 ($\chi^2(1) = 4.184$, $p < .05$) but not between model 1 and 2 ($p > .05$). Meeting centralization had a negative significant relationship with team performance. By contrast, participation equality and individual participation were not significantly linked to team performance. Thus, meeting centralization was a better measure of team heterogeneity in the meetings and a better predictor of team performance than participation equality.

The results demonstrate the importance of decentralized team meeting interaction for perceived team performance. The sheer quantity of individual contributions by contrast has no impact on participants' team performance rating. Thus, assessment of perceived team performance is related to the overall interaction structure and not to individual participation in the meeting. Beyond this effect, our research sample also shows substantial range in network

centralization. We found network centralization to be evenly distributed between the highest and lowest centralized team meeting. This leads to a number of research questions. Why to some teams tend to form more centralized interaction structures than others? Can we identify potential antecedents that predict the degree to which meeting participants will be involved in the meeting interaction process? Different moderators might also play an important role in this context. For example, homophily (beyond gender and age) might have a strong impact on meeting centralization. Participants' educational background, team tenure, or friendships within the group might affect the extent to which they engage in the meeting process and whether they form subgroups. Moreover, meeting design characteristics such as different meeting agendas might influence meeting interaction and lead to different network structures. All in all, we can conclude that more research on interaction networks in team meetings is needed to gain a deeper understanding of the coordination and structure in team meetings. We provide a long list of open research questions at the end of our chapter that hopefully triggers researchers' interest in studying network structures in team meetings.

Challenges for Meeting Research

Besides the manifold opportunities the network perspective offers to meeting research, there are several challenges researchers need to consider when applying social network analysis to meeting interaction data. First of all, the whole meeting process needs to be captured; preferably using video recordings. In doing so, researchers need to be mindful of the video quality as background noises can interfere with the discussion and hamper later analyses. Even more importantly, it is crucial for researchers to build trust with all meeting participants. Being videotaped in a work setting is an unfamiliar situation for most meeting participants. Moreover, it entails that all spoken words are saved and thus 'leave the room' which can cause uncertainty. Meeting participants might fear that the recordings could be used against them by superiors or other members of the organization at later points in time. It

is thus important to agree by contract that the researchers are the only people who have access to the video recordings. Research shows that meeting participants quickly get accustomed to being videotaped and qualify videotaped meetings as typical as under usual circumstances (e.g., Kauffeld & Lehmann-Willenbrock, 2012).

Another challenge is the coding of the data as coders need training. Cutting the stream of speaking-turns from the video sounds like an easy task at first. However, coders need a thorough set of coding rules for identifying speaking-turns to obtain reliable results. For example, the meeting researcher must decide whether statements such as “Uh-Huh” are a counted as a speaking-turn or not. Another problem surrounds the distinction between outdegree (i.e., number of times an actor addresses other participants) and indegree (i.e., number of times an actor responds to previous utterance) in interaction networks based on speaking-turns. In some cases, meeting participants address their utterances to the group as a whole and not to one specific meeting participant. However, it is not always possible to account for such cases. Also, participants might react to a previous utterance even though this utterance was not intentionally directed at them (e.g., someone answers a question for someone else). Feeding speaking turns into a “who-to-whom”-matrix, an utterance is at the same time counted as an outdegree of the current speaker and as an indegree of the meeting participant who reacts to the same utterance and picks up the flow of verbal meeting interaction. We advise meeting researchers interested in social network analysis based on speaking-turns to carefully choose and interpret their network measures.

Finally in applying the network perspective to meeting research, it is important to be aware of the data’s specific features. The interaction networks are valued and directed graphs. Meetings also differ in their sum of speaking-turns. The network measures to be calculated thus need to be chosen with caution. As social network analysis is mostly focused on binary data, a wide range of network measures do not fit the data specifications or cannot be

calculated for valued digraphs by specific computing programs. However, big steps are currently made in developing software packages for the calculation of valued digraphs including valued exponential random graph models that offer new possibilities of simulation analyses of the interaction networks (Krivitsky, 2012).

Benefits for Meeting Research and Future Research Avenues

In the application sample described above, we focused on the overall structure of the interaction network and its link to team outcomes. Applying a network perspective to meetings can have several important benefits. First of all, the wide array of techniques introduced above offer meeting researchers an entirely new depth of analysis that goes beyond the quantity of meeting participation. Meeting researcher interested in using social network analysis can gain additional insights into the dynamics of the meeting process by taking into account that all meeting participants are interdependent. As we described before, the conversational responses shape the team interaction in meetings and can thus be used to construct an interaction network. In the following, we will highlight different benefits of using social network analysis on team meeting interaction data and introduce research questions that yet need to be answered. Table 6 provides an overview of different benefits and lists possible research questions.

Insert Table 6 about here

Focus on position: On the individual level, actors' positions and roles can be identified and the influence of particular actors can be examined. Researchers have found that attributes such as gender, age, and status affect actors' participation in small groups (Bonito & Hollingshead, 1997). From a network perspective, these findings can be explored in more detail. Do attributes such as high status, age, or tenure lead to a position as a connecting link

with high betweeness and thereby put the participant in the position of facilitator? Researchers also found that participants with high quantities of participation are perceived as more influential (Correll & Ridgeway, 2003). Network analysis can be used to build on these findings, examining the impact of different aspects of centrality. Are actors with high numbers of direct ties perceived as meeting leaders? Are actors with high betweeness scores who broker interaction between unconnected meeting participants perceived as facilitators? Beyond central actors, isolates can be identified and their impact on meeting effectiveness and meeting satisfaction can be examined. In a recent network study, Sauer and Kauffeld (subm.) explored the roles of meeting leaders by applying network analysis to meeting interaction. Meeting leaders took on the roles of central actor with a higher degree centrality, broker with a higher betweeness, and elicitor with a higher assortativity than regular meeting participants. The position of central actor had a positive effect on meeting effectiveness whereas the position as elicitor had a positive impact on meeting satisfaction and team productivity after the meeting. Furthermore, the network perspective offers the opportunity to check if typical network effects can be found in meeting interaction as well. Does reciprocity have an influence on the interaction so that actors tend to discuss topics in dyads? Do central actors also have high assortativity scores mostly interacting with other participants central to the discussion? Can actors be identified who take on brokerage roles by connecting different subgroups and fostering interchange in team meetings?

Focus on subgroups: As described above, a key variable influencing the development of networks is *homophily* which describes the tendency of individuals to bond with similar others (Higgins, Chandler, &Kram, 2007). Attributes affect the formation of network patterns, which causes gendered subgroups to frequently emerge in work teams (Ibarra, 1992).This effect needs to be examined in meetings. Are men closely connected to other men and women tightly linked to other women in the interaction networks? Do status and age influence the

interaction structure so that older, higher-status participants are located in the core and younger, lower-status participants situated in the periphery? How does subgroup formation influence the functioning of the group? Triads are called the building block of networks (Wasserman & Faust, 1994). A typical network phenomenon is called *transitive triads*, which display equilibrium between the three involved actors as all three have balanced connections. Can triads be identified as typical patterns in meeting interaction? How do different types of patterns (*transitive triads* vs. *structural holes* vs. *clusters*) influence group functioning?

Focus on overall structure: Research on communication networks has found that, depending on task complexity, the centralization of interaction has an influence on problem solving and group productivity (e.g., Bavelas 1950; Crawford & LePine, 2013; Sparrowe et al., 2001). Sauer and Kauffeld (2013) used the network approach to demonstrate that centralized meeting interaction has a negative impact on team performance. Another research area that might highly benefit from a network perspective is intercultural differences in team meeting interaction. Lehmann-Willenbrock, Allen, and Meinecke (2014) recently showed that German and U.S.-American team meeting behaviors significantly differ. Moreover, a recent qualitative study (Köhler, Cramton, & Hinds, 2012) showed that Germans and U.S.-Americans have different norms and expectations about appropriate communication and coordination behaviors in team meetings. U.S.-Americans were said to value an impromptu and linear meeting style. Germans, on the other hand, preferred more structured meetings and repeatedly returned to and refocused on task definition (Köhler et al., 2012). To date, we do not know how these findings translate to the coordination and structure in team meetings. For example, do Germans prefer more centralized team meeting structures than U.S-Americans?

Focus on content: Thus far, we have treated meeting interaction solely from a structural perspective. However, there are further features of meeting interaction that can be assessed by means of social network analysis. One opportunity is to take the content of the

interaction into account. A quite simple distinction could be made by separating functional and dysfunctional meeting interaction (see Meinecke & Lehmann-Willenbrock in this volume). In this case, functional interaction would comprise all the positive utterances such as problem- and solution-focused, procedural or action-oriented statements whereas dysfunctional interaction would comprise negative utterances such as complaining or criticizing. On the basis of this distinction, the overall structure of functional vs. dysfunctional interaction networks could be compared. Are these structures associated and how do they influence group functioning? A decentralized functional interaction network could cause centralized dysfunctional interaction with isolated complainers and result in high meeting satisfaction. Are specific positions in the different networks related? A facilitator with a high centrality in the functional network could have a decentralized position in the dysfunctional network, whereas a bad apple with a high centrality in the dysfunctional network could be an isolate in the functional network.

Focus on context: The network perspective could also help meeting researchers examine phenomena that are difficult to pin down and have received little attention thus far (Katz et al., 2004). The influence of the social situation on meeting participants' behavior during the meeting could be such a phenomenon. As argued above, researchers could analyze in which way multiplex relationships between the participants outside the meeting (e.g., friendship, support, or advice) influence the meeting interaction. Do friends interact more closely in meetings? Are employees central in the advice network also central participants in the team meeting? Beyond the internal ties of meeting participants, the external ties of the group should be explored. Group scholars have long called for more attention to groups' external environments (Ancona & Caldwell, 1992; Stohl & Putnam, 2003). The external ties might emanate from individual meeting participants or the whole team and might connect to individual employees and teams that are part of the organization or individual partners and

groups from outside the organization. Does a well-connected team have a high transactive memory, which influences the meeting interaction and results in higher meeting effectiveness? Do connections to high-status outsiders provide an actor with power, which results in a central position in the interaction network?

Focus on dynamics: Lately, big steps have been made to study network dynamics (Snijders, 2013; Snijders & Steglich, in press). This offers a wide array of opportunities for meeting researchers as it allows the analysis of emerging network patterns over time. Studies on meeting design characteristics (e.g., Leach, Rogelberg, Warr, & Burnfield, 2009) identified important characteristics that are related to perceived meeting quality. For example, setting up an agenda at the beginning of the meeting leads to higher perceptions of meeting quality. Investigating the development of network patterns over different meeting phases should extend our knowledge about the specific use of meeting designs and meeting facilitation. Are meetings effective that are centralized at the beginning and end of the meeting but decentralized in the middle? Investigating the emergence of interaction patterns is also interesting from a leadership point of view. How do meeting leaders emerge over the course of the meeting? Are meetings successful when appointed leaders take on a central role at the beginning and the end when setting the agenda and summarizing the results but occupy a decentral position during the discussion when focusing on facilitating interchange?

Practical Implications for Meetings

The approach we presented in this chapter challenges meeting researchers to broaden their perspective on the meeting process. However, the application of social network analysis to meeting interaction not only offers scholars a wide range of tools to complement the traditional approach to meetings, it also provides opportunities for managers and team members to understand configurations of meeting interaction. With simple tasks such as allocating work, centralized interaction is beneficial as it lessens participants' communication

burdens. Information can be channeled more effectively and redundancy can be avoided (e.g., Shaw, 1964; Crawford & LePine, 2013). With complex tasks such as finding solutions to current work problems, however, decentralized interaction fosters meeting effectiveness with all participants sharing their knowledge (e.g., Huang & Cummings, 2011; Sauer & Kauffeld, 2013). Team training could thus be helpful to make meeting leaders and participants aware of the pros and cons of different interaction structures. Research on sociometric data has shown that visual interaction maps are helpful for teams to identify their interaction patterns (Pentland, 2012). These visualizations could be used as a tool in team trainings to diagnose how to facilitate functional meeting interaction and diminish counterproductive meeting behavior.

Recording and visualizing the interaction in meetings could be used in multiple ways beyond team trainings. Managers could derive implications about effective meeting designs. Visualizing the interaction structure in team meetings could provide information if a facilitator is needed to direct the meeting process or if the interaction thrives without a designated meeting leader. Following research on the effect of homophily on communication ties (Brass, 2012), the analysis of who interacts with whom in meetings could also give indications on team composition. Managers could detect if age diversity fosters or hampers the interaction and if interaction patterns are characterized by gender. Moreover, meeting roles such as the ‘bad apple’ (Felps, Mitchell, & Byington, 2006) could be identified that promote or diminish meeting effectiveness. As a result, areas offering potential development opportunities could be detected.

Along these lines, mapping team members’ multiple external relations could provide fruitful insights into the team’s position in the organizational structure. Team members occupying broker positions with ties to other teams in the organization could be identified,

relationships to unconnected teams could be built, and existing ties could be diversified to purposefully increase the team's organizational knowledge.

Conclusion

Research on networks in small groups started with promising results in the 1950s but experienced a decline until the 1990s. Over the last 20 years, encouraging studies have been conducted on communication networks in real work groups that recently resulted in the application of social network analysis on team meeting data. We believe that social network analysis offers meeting researchers an exciting new approach to study meetings. It entails a new perspective on the meeting process focusing on the structural aspects of meeting interaction that have been neglected thus far. Social network analysis allows researchers to gain traction on a great deal of interaction phenomena.

On the individual level, the position and role of meeting participants and meeting leaders can be examined. On the group level, overall meeting structures can be explored. Examining the antecedents and consequences of interaction patterns gives scholars the opportunity to broaden their knowledge about meeting functioning and facilitation. Beyond the examination of internal meeting interaction, the context and environment of the team meeting can be taken into account as participants' ties to employees from other teams or other organizations can be included. Analyzing ties from individuals to groups and groups to organizations allows meeting researchers to get a grip on cross-level phenomena that might influence the meeting process. Two interesting areas of network research deserve particular attention. First, social network analysis has developed tools to analyze multiplex network ties. Social relations among employees in organizations are usually complex encompassing multiple kinds of relationships such as being friends, giving advice, or being a mentor. These simultaneous relations shape the interaction between team members in meetings. Second, the longitudinal analysis of network data has only just begun and offers broad opportunities to

track the emergence of specific interaction patterns among participants, the formation of subgroups, the change of positions in the network, or the development of the interaction structure.

In sum, first steps have been taken to conceptualize meeting interaction as networks. Now, more research is needed to tap the full potential of using social network analysis on team meeting interaction data.

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Table 1

Typology of Ties

Similarities			**Social Relations**				**Inter-action**	**Flows**
Location	Member-ship	Attri-bute	Kinship	Other Role	Affective	Cognitive		
e.g., same spatial & temporal space	e.g., same clubs, same events	e.g., same gender, Same attitude	e.g., mother, sibling	e.g., friend, boss, student	e.g., likes, hates,	e.g., knows about, sees as happy	e.g., talked to, gives advice to	e.g., infor-mation, beliefs, personnel, resources

(Borgatti, Mehra, Brass, & Labianca, 2009)

Table 2

Relational Social Network Measures of Ties

Measure	**Definition**	**Example**
Direction	Extent to which ties go from one actor to the other	Non directional: being a member of a group; Directional: giving advice to a team member
Medium	How contact is established	Face-to-face contact, mail contact etc.
Value	Extent to which the relationship is scaled	Binary: being teammates; Valued: being friends on a scale of one to ten
Frequency	How often the tie occurs	Daily chats, weekly meetings, etc.
Strength	Degree of intensity	Strong ties: high emotional intensity or intimacy; often seen in relationships to family and friends; Weak ties: low emotional intensity or intimacy; connections with acquaintances
Valence	Positive vs. negative aspects of relationships	Positive: benefits of actors' relationships; Negative: liabilities of actors' relationships
Multiplexity	Extent of ties between actors across different types of relationships	Team members are connected by friendship and advice ties

Table 3

Social Network Measures for Individual Actors

Measure	**Definition**	**Relation to Studying Meetings**
Centrality	An actor's structural importance in the network	An actor's contribution to the meeting interaction
Degree	Actor's number of direct ties to others	An actor's direct influence in the meeting
Indegree	Number of in-coming ties (towards an actor)	How many times an actor responds to a previous utterance
Outdegree	Number of out-going ties (away from an actor)	How many times an actor addresses other participants
Betweeness	Extent to which an actor mediates interchange between other members	An actor's control over the information flow in the meeting
Closeness	Extent to which an actor reaches all others	How many speaking-turns it takes for an actor to respond to another participant in the meeting
Assortativity	An actor's preference to connect with similar others	Tendency of a central actor to interact with other central participants in the meeting
Effective size	Extent of structural holes in an actor's network	How much the meeting is focused on the dyadic interaction of specific actors

Table 4

Social Network Measures for Subgroups

Measure	**Definition**	**Relation to Studying Meetings**
Components	Groups in which all actors are either directly or indirectly connected with each other and no actors have ties outside the component	Groups exclusively interacting with each other during the meeting
Cliques	Complete subgroups in which all actors are directly connected with each other	Subgroups in which all participants heavily interact with each other
Structural equivalence	Actors are clustered based on similar patterns of ties, without necessarily being directly linked	Subgroups with equivalent interacting patterns in the meeting

Table 5

Social Network Measures for Entire Networks

Measure	**Definition**	**Relation to Studying Meetings**
Centralization	Degree to which a network revolves around a single actor	Degree to which all participants are involved in the meeting interaction
Distance	Cohesion within the network	Probability of two participants interacting with each other
Coreness	Extent to which a network is structured around a core of closely linked members	How much is the interaction focused on a core of heavily interacting participants
Core-Periphery Model	Actors position in the core vs. periphery of the network	Which are active participants positioned in the core of the meeting interaction and which are detached participants positioned in the periphery
Transitivity	Extent to which connected actors are linked to the same others	How much dyadic and triadic interaction occurs in the meeting

Table 6

Benefits for Meeting Research

Focus on	**Benefits**	**Possible Research Questions**
Position	Examining antecedents and consequences of actors' positions and roles in the meeting	Which attributes characterize central actors in meetings? Which participants take on brokerage roles by connecting different subgroups and fostering interchange in meetings?
Subgroups	Examining hubs of interaction in the meeting	Are there homophily effects in meetings (gender, education etc.)? How does subgroup formation influence the functioning of the team?
Overall structure	Examining antecedents and consequences of different interaction structures in the meeting	Which interaction structures promote meeting effectiveness and satisfaction? Are their cultural differences in the coordination of meetings? How does the meeting design influence the interaction structure in meetings?
Content	Examining different types of relationships in the meeting	How does the structure of functional and dysfunctional interaction differ in meetings? Are there 'energizers' (actors positioned in the functional core but dysfunctional periphery) and 'bad apples' (actors positioned in the functional periphery but dysfunctional core) in meetings?
Context	Integrating the team environment beyond the meeting	Is friendship influencing participants' interactions in

		meetings? Are ties to external teams and partners influencing the interaction in the meeting?
Dynamics	Examining the emergence of interaction patterns over time	Does the interaction structure evolve over the course of the meeting? How do meeting leaders emerge over the course of the meeting?

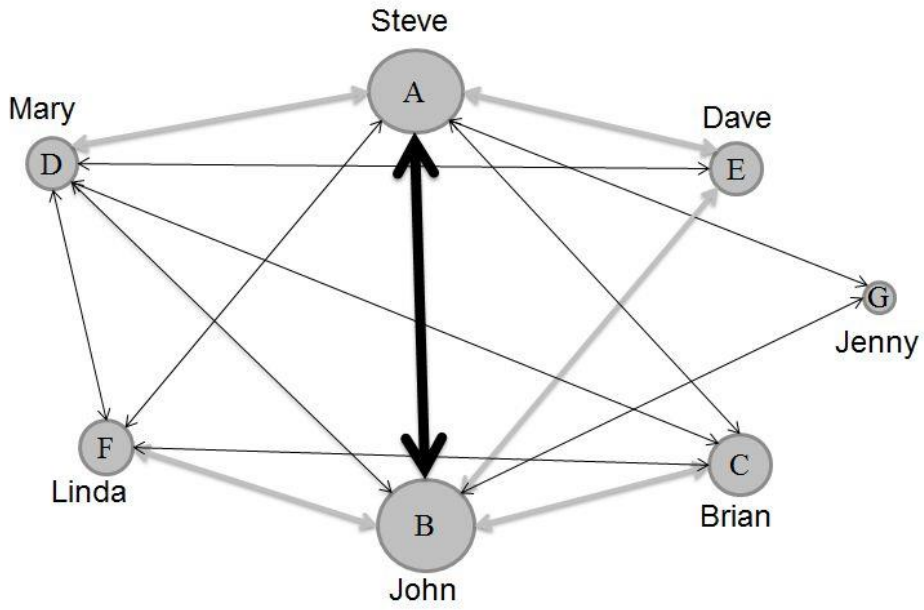

Figure 1.Weighted graph of meeting interaction in a complete team meeting. Nodes display participants. Arcs display participants' responses.

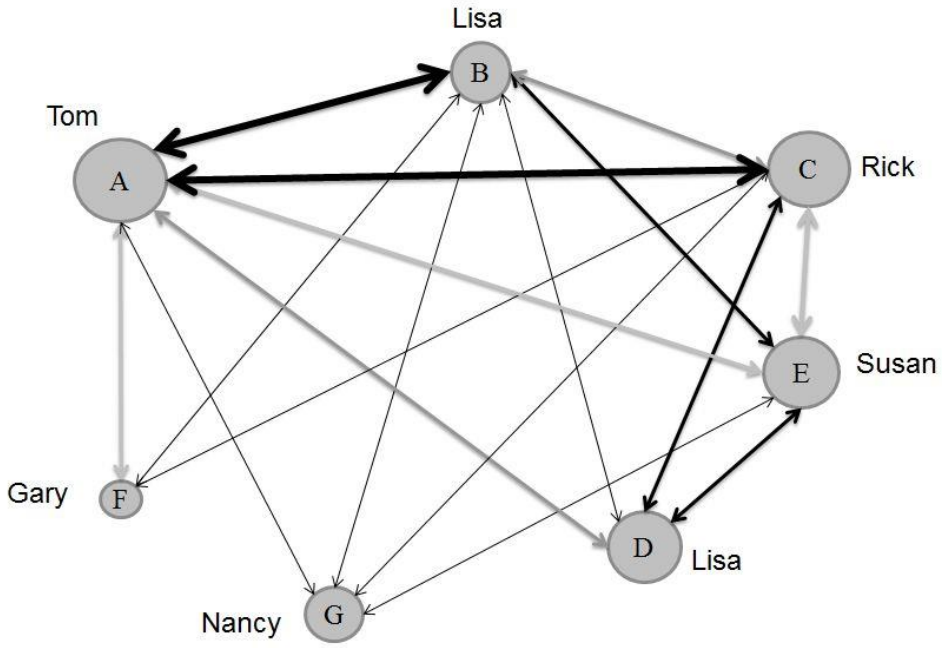

Figure 2. Weighted graph of meeting interaction in a complete team meeting. Nodes display participants. Arcs display participants' responses.

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